

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Subject: 394173 Mathematics III

Date: 29 July 2013

Year: 2013

Section: 15-16

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. **No** documents are allowed to be taken out of the examination room.
3. Textbooks and dictionary are **NOT** allowed.
4. Calculators are allowed in the examination.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 9 pages (including this page), 10 questions and 1 part.
The total score is 72 points.
7. Write all your answers on this examination sheet.

No.	score
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
9.	
10.	
Total	

1. Let $z_1 = 2 + 3i$, $z_2 = -4 + 4i$ and $z_3 = -3 + i$. Find

1.1 $z_1 + 2\overline{z_3}$

1.2 $z_2 \bullet z_1$

1.3 $|(z_3 - z_1)^4|$

1.4 Write the polar form of z_2 .

(12 points)

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2. Assume that $z_1 = \frac{4i}{1-i^{555}}$ and $z_2 = \frac{7i^{88} + 24i^{33}}{5i}$. Evaluate $\left| \frac{\overline{z_1}(z_2)^2}{z_1} \right|$.

(7 points)

3. Find the sum of solutions to $z^5 - z^4 - 8z^2 + 8z = 0$.

(6 points)

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4. Let z_0, z_1, z_2 and z_3 be 4th roots of $-1 - \sqrt{3}i$, find $\left| \frac{z_1 \cdot z_3}{z_0 \cdot z_2} \right|$. (7 points)

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5. Given that $\frac{3}{2} \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 10 & 6 \\ -8 & 7 \end{bmatrix} - 3 \begin{bmatrix} 3 & 2 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$

Determine the value of $a + b + c + d$.

(6 points)

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6. Evaluate A^{-1} , where $\begin{bmatrix} 5 & -1 & 2 \\ -4 & 1 & -2 \\ 3 & 0 & 1 \end{bmatrix} \bullet A = \begin{bmatrix} 0 & 3 & 2 \\ -1 & 2 & 3 \\ 4 & -2 & -3 \end{bmatrix}$. (7 points)

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7. Assume that A , B , C and I_2 be matrices 2×2 , where I_2 is the identity matrix.

If $\det(2\sqrt{2}A) = 4$, $\det(C^{-1}) = 4$ and $AB^TC = \begin{bmatrix} -6 & 1 \\ 4 & -1 \end{bmatrix}$, find $\det(B)$?

(7 points)

8. Solve the system of equations using Gauss–Jordan elimination.

(6 points)

$$\begin{aligned} x_1 + x_2 - 3x_3 &= 4 \\ x_2 + x_3 &= 1 \\ 3x_1 + 2x_2 - x_3 &= 5. \end{aligned}$$

9. Solve x_3 , using Cramer's rule:

(6 points)

$$x_1 + x_2 = 0$$

$$x_2 + x_3 - 2x_4 = 1$$

$$x_1 + 2x_3 + x_4 = 0$$

$$x_1 + x_2 + x_4 = 0.$$

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10. Kook-Kai Shop makes bookshelves and tables. Each bookshelf requires 5 hr. of woodworking and 4 hr. of finishing. Each table requires 10 hr. of woodworking and 3 hr. of finishing. Each month the shop has 600 hr. of labor available for woodworking and 240 hr. for finishing. The profit on each bookshelf is \$40 and on each table is \$75. How many of each product should be made each month in order to maximize profit? (8 points)

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